Multi-Peak Ground-State Solutions for the Nonlinear Schrödinger Equation with External Potential

Eduard Kirr
Department of Mathematics
University of Illinois at Urbana-Champaign

Ethan Knox
Department of Mathematics
University of Illinois at Urbana-Champaign

March 2026

Abstract

We classify multi-peak ground-state configurations of the stationary nonlinear Schrödinger equation $(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0$ in the high-energy regime $E \to \infty$. Using the rescaling and concentration compactness framework of Kirr and Natarajan [7], ground states decompose into well-separated profiles whose positions are governed by a finite-dimensional algebraic system balancing the Hessian $H_V(x_0)$ of the potential against nonlinear inter-peak interactions.

We prove that near a non-degenerate local maximum x_0 of V, the admissible multi-peak configurations are: collinear arrangements along a single eigenvector of H_V; isosceles triangles requiring the eigenvalue constraint $\lambda_2 = 3\lambda_1$; equilateral triangles requiring $|\lambda_2|/|\lambda_1| \geq 3$; and equilateral triangles with a continuous rotational degree of freedom θ, admissible for all orientations when $\frac{1}{3} \leq \mu \leq 3$ (where $\mu = |\lambda_2|/|\lambda_1|$) and for restricted angular windows otherwise. To extend these to finite energy, we construct explicit solutions of the perturbation system $F_i(\delta y, R) = 0$ at $R = 0$, identify the four-dimensional kernel of the linearization, and show that gauge-fixing constraints reduce the problem to a system amenable to the implicit function theorem.

Keywords: nonlinear Schrödinger equation, multi-peak solutions, concentration compactness, Lyapunov–Schmidt reduction, ground states, optical fibers

MSC 2020: 35Q55, 35B40, 35B32, 78A60

Contents

1 Introduction **2**

2 Setup and Preliminaries **3**
2.1 Hypotheses on the potential . . . 3
2.2 Rescaling and concentration compactness . . . 5
2.3 Lyapunov–Schmidt reduction . . . 8
2.4 The reduced algebraic system . . . 8

3 Classification of Limiting Configurations **9**
3.1 Collinear configurations . . . 9
3.2 Isosceles triangle . . . 11
3.3 Equilateral triangle . . . 11
3.4 Rotational equilateral triangle . . . 11
3.5 Admissibility analysis . . . 11

4 The Perturbation System **13**
4.1 Formulation . . . 13
4.2 Explicit solution at $R = 0$ . . . 13
4.3 Kernel of the linearization . . . 15

5 Kernel Resolution and Gauge-Fixing **15**
5.1 Projection decomposition . . . 16
5.2 Gauge-fixing constraints . . . 16
5.3 Rotational equivariance at $R = 0$ . . . 16
5.4 Extension to $R > 0$: invertibility of the gauge-fixed system . . . 16

6 Conclusion and Open Problems **17**

A Notation Reference **18**

1 Introduction

The propagation of intense light through optical fibers is governed by the interplay between linear dispersion and the nonlinear response of the dielectric medium. In silica, the centrosymmetric lattice eliminates second-order nonlinearity, so the leading nonlinear effect is the Kerr term $\chi^{(3)}$. Starting from Maxwell's equations in a source-free, non-magnetic dielectric, the electric field can be written as a slowly varying envelope modulating a carrier:

$$\mathbf{E}(\mathbf{r}, t) = \tfrac{1}{2}\hat{\mathbf{x}}\big[F(x, y)\, A(z, t)\, e^{i(\beta_0 z - \omega_0 t)} + \text{c.c.}\big].$$

Here $F(x, y)$ is the transverse modal shape set by the fiber's refractive-index profile, $A(z, t)$ is the envelope, and β_0 is the propagation constant at the carrier frequency ω_0. Applying the slowly varying envelope approximation and expanding the propagation constant to second order in frequency yields the nonlinear Schrödinger equation (NLS):

$$i\partial_z A - \tfrac{\beta_2}{2}\partial_T^2 A + \gamma|A|^2 A = 0,$$

where β_2 is the group-velocity-dispersion parameter and $\gamma = n_2\omega_0/(cA_{\text{eff}})$ encodes the Kerr nonlinearity.

In real fibers the refractive index is not spatially uniform. Fiber Bragg gratings impose periodic modulations, graded-index fibers create a parabolic transverse profile (an effective harmonic-oscillator potential), and photonic-crystal fibers achieve confinement via a micro-structured air-hole lattice. In every case, the spatial variation of the index enters the governing equations as an external potential $V(\mathbf{x})$. The transverse modal equation

$$\nabla_\perp^2 F + \Big[\big(\tfrac{n\omega}{c}\big)^2(\mathbf{x}_\perp) - \beta^2\Big]F = 0$$

is mathematically identical to the stationary Schrödinger equation $-\Delta\psi + V(x)\psi = E\psi$. Guided modes of the fiber correspond to bound states of this effective potential, and the fundamental mode is its ground state. This equivalence motivates the present work: engineering $V(\mathbf{x})$ via refractive-index design directly translates into questions about the existence, uniqueness, and stability of ground-state solutions of the NLS with an external potential.

Consequently, we study the stationary NLS

$$(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0, \tag{1.1}$$

with $V : \mathbb{R}^n \to \mathbb{R}$, $\psi : \mathbb{R}^n \to \mathbb{C}$, $\sigma > 0$, and $E \geq 0$. The parameter E represents the energy (or, in the optical setting, the propagation constant). For each admissible E we seek ground-state solutions ψ_E and analyze how their spatial structure is governed by the geometry of V.

Since (1.1) is posed on the unbounded domain $\mathbb{R}^n$, compactness is lost and standard variational methods fail. The concentration compactness principle introduced by Lions [9] classifies minimizing sequences into compactness, vanishing, or dichotomy. Using the rescaling of Kirr and Natarajan [7], one obtains a bounded family in H^1. Concentration compactness then yields a multi-peak decomposition: u_E is approximated by a finite sum of well-separated profiles, each solving the translation-invariant NLS. The rescaled peak positions converge to the critical points of V as $E \to \infty$.

This multi-peak decomposition reduces the infinite-dimensional problem to a finite-dimensional system governing the peak locations. A Lyapunov–Schmidt reduction first solves for the orthogonal remainder and then projects onto the span of the translational modes, yielding an algebraic system that balances the restoring force from $H_V(x_0)$ against the attractive interaction between neighboring peaks. The overall analytical pipeline, building on [7] and extended in the present work, is illustrated in Figure 1.

Our contributions are threefold. First, we systematically classify all admissible multi-peak configurations near a non-degenerate local maximum of V (Theorem 8). Second, we formulate and solve the perturbation system at $R = 0$ (Proposition 12), identify the kernel obstruction (Proposition 13), and propose a gauge-fixing strategy for its resolution (Section 5). Third, we provide explicit admissibility regions in the (μ, θ) parameter space for equilateral triangle configurations with rotational freedom.

The multi-peak construction builds on a long line of work on semiclassical and spike solutions of NLS. Floer and Weinstein [1] first proved the existence of standing waves concentrated near nondegenerate critical points of V for bounded potentials. The concentration compactness framework for multi-peak decomposition was developed in [9, 10], and the symmetry-breaking bifurcation analysis for NLS with symmetric potentials was carried out in [5]. Stability of solitary waves under symmetry was established by Grillakis, Shatah, and Strauss [2, 3], and asymptotic stability of ground states was proved in [8, 6]. The present work extends the global bifurcation analysis of [7] by providing the first complete geometric classification of multi-peak configurations and a systematic resolution strategy for the kernel obstruction in the reduced system.

2 Setup and Preliminaries

2.1 Hypotheses on the potential

We impose the following conditions on V, following [5, 1, 7]:

Hypothesis V1. $V \in L^\infty(\mathbb{R}^n)$, i.e., V is bounded and measurable.

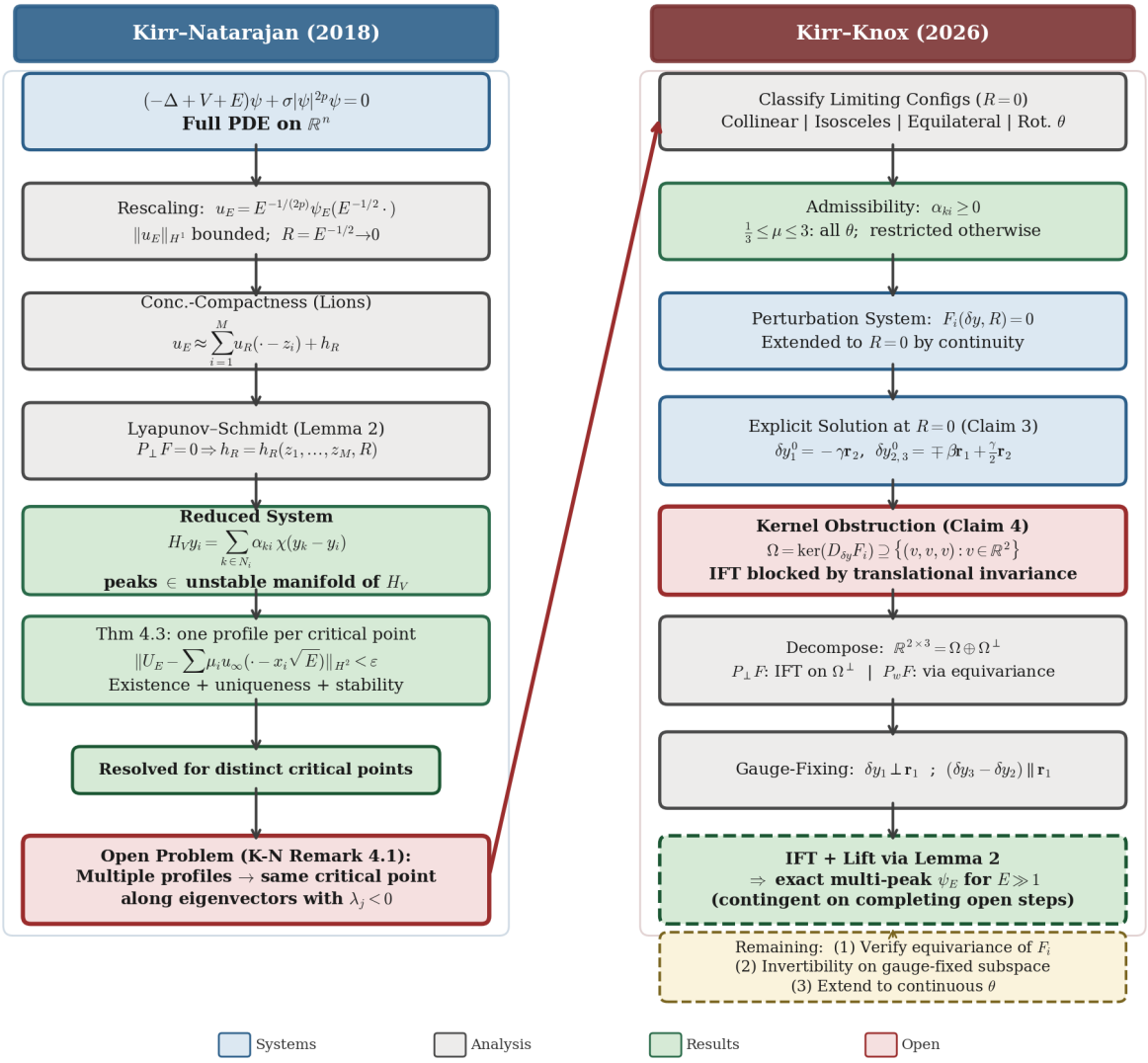

Figure 1: Analytical pipeline from the PDE to multi-peak ground states. The left column shows the framework established in Kirr–Natarajan [7]: rescaling, concentration compactness decomposition, and Lyapunov–Schmidt reduction to a finite-dimensional algebraic system. The right column shows the contributions of the present work: classification of admissible peak geometries, construction of the perturbation system $F_i(\delta y, R) = 0$, kernel analysis, and gauge-fixing strategy.

Hypothesis V2. $\lim_{|x|\to\infty} V(x) = 0$.

Hypothesis V3. V is C^2 in a neighborhood of x_0, where x_0 is a non-degenerate critical point of V: $\nabla V(x_0) = 0$ and $H_V(x_0)$ is invertible.

Hypothesis V4. All eigenvalues of $H_V(x_0)$ are strictly negative, i.e., x_0 is a strict local maximum of V. We write $\lambda_1, \ldots, \lambda_n < 0$ for the eigenvalues with corresponding orthonormal eigenvectors $\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n$.

Hypothesis V1 ensures that $L_0 = -\Delta + V$ is self-adjoint on $L^2(\mathbb{R}^n)$ with domain $H^2(\mathbb{R}^n)$. Together with V2, it guarantees that the essential spectrum of L_0 is $[0, \infty)$ and that any negative eigenvalues are discrete. Hypothesis V3 is the minimal regularity needed for the Taylor expansion underlying the Lyapunov–Schmidt reduction. Hypothesis V4 is the structural condition that forces multi-peak configurations to concentrate at x_0; peaks at a local minimum would correspond to $\lambda_i > 0$, for which the algebraic system has no admissible solutions with $\alpha_{ki} \geq 0$.

Remark 1. Floer and Weinstein [1] work with V bounded and prove concentration near any nondegenerate critical point (maximum or minimum). For the multi-peak case, the requirement $\lambda_i < 0$ arises from the interaction analysis: the peaks must lie in the unstable subspace of $H_V(x_0)$ (Corollary 7).

2.2 Rescaling and concentration compactness

Following [7], define

$$u_E(x) = E^{-\frac{1}{2p}}\, \psi_E(E^{-\frac{1}{2}} x), \tag{2.1}$$

so that $\|u_E\|_{H^1}^2 = \|\nabla \psi_E\|_{L^2}^2 + \|\psi_E\|_{L^2}^2$ remains bounded. By the concentration compactness principle [9], there exists $M \in \mathbb{N}$ such that

$$\left\| u_E(x) - \sum_{i=1}^{M} u_i(x - z_i(E)) \right\|_{H^1} \xrightarrow{E\to\infty} 0, \tag{2.2}$$

where each profile $u_i : \mathbb{R}^n \to \mathbb{C}$ solves the translation-invariant equation

$$(-\Delta + 1)u + \sigma |u|^{2p} u = 0. \tag{2.3}$$

Setting $R = E^{-1/2}$, the rescaled NLS becomes

$$(-\Delta + R^2 V(Rx) + 1)\, u_E + \sigma |u_E|^{2p} u_E = 0. \tag{2.4}$$

Under Hypotheses V1–V2, the rescaled peak positions $Rz_i(R)$ remain bounded and converge to the critical points of V as $R \to 0$.

Proposition 2 (Kirr–Natarajan [7]). *Under Hypotheses V1–V4, $Rz_i(R) \to x_0$ for each $i = 1, \ldots, M$ as $R \to 0$.*

We rewrite (2.2) as

$$u_E(x) = \sum_{i=1}^{M} u_R(x - z_i(R)) + h_R(x),$$

where u_R is the unique (up to phase) positive, spherically symmetric solution of $(-\Delta + R^2 V(x_0) + 1)u + \sigma |u|^{2p} u = 0$.

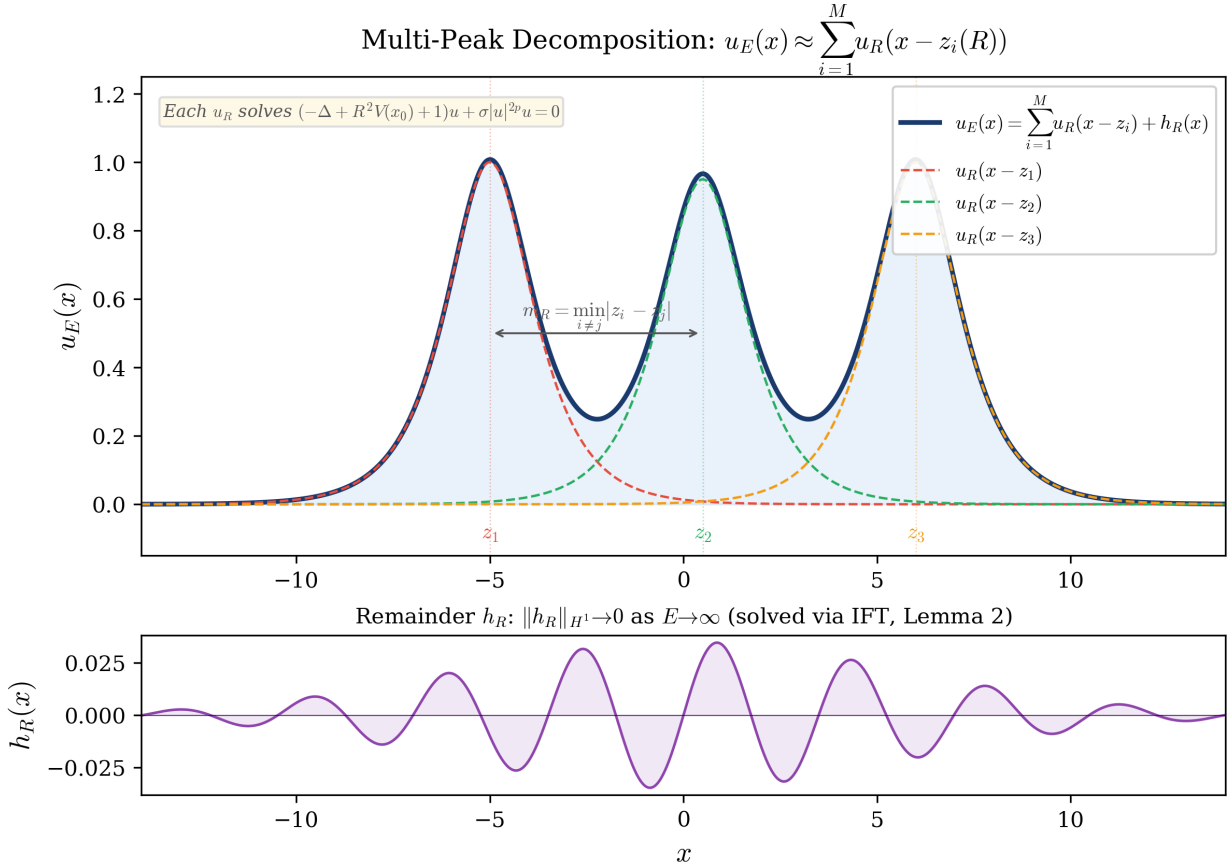

Figure 2: Multi-peak profile decomposition. The rescaled solution u_E (solid black) is approximated by a sum of well-separated profiles $u_R(\cdot - z_i)$ (colored dashed), each solving the translation-invariant NLS (2.3). The remainder h_R (gray shaded) satisfies $\|h_R\|_{H^1} \to 0$ as $R \to 0$.

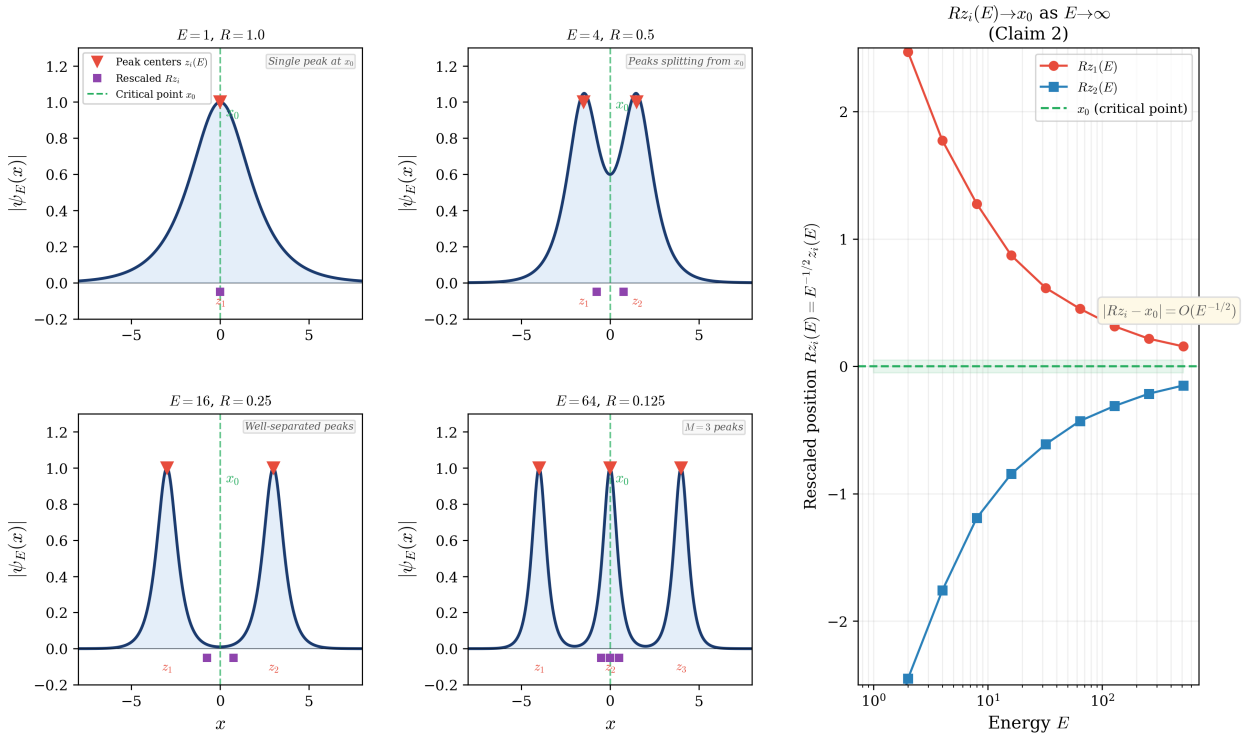

Figure 3: Concentration of rescaled peak positions. As $E \to \infty$ (equivalently $R \to 0$), the rescaled positions Rz_i converge to the critical point x_0 of V. The peak positions z_i in the blown-up coordinates separate to distances $\sim m_R$, the minimum inter-peak spacing.

2.3 Lyapunov–Schmidt reduction

The derivatives $\partial u_R/\partial x_j$ satisfy the linearized equation and generate the translational kernel. The Lyapunov–Schmidt reduction proceeds in two steps:

Lemma 3 (Lyapunov–Schmidt, Step 1 [7]). *There exists $\varepsilon > 0$ such that for $0 < R < \varepsilon$, there exist adjusted centers $\tilde{z}_1, \ldots, \tilde{z}_M \in \mathbb{R}^n$ with $u_E(x) = \sum_{i=1}^M u_R(x - z_i(R)) + \tilde{h}_R(x)$ and $\tilde{h}_R \perp \partial_{x_j} u_R(\cdot - z_i(R))$ for all $i = 1, \ldots, M$ and $j = 1, \ldots, n$.*

Lemma 4 (Lyapunov–Schmidt, Step 2 [7]). *There exist $\tilde{\varepsilon}, \delta, d > 0$ such that if $R < \tilde{\varepsilon}$, $\|h\|_{H^1} < \delta$, and $|z_i - z_j| > d$ for $i \neq j$, then $P_\perp F(h, z_1, \ldots, z_M, R) = 0$ has a unique solution $h = h(z_1, \ldots, z_M, R)$.*

2.4 The reduced algebraic system

Projecting onto the translational modes yields the reduced system. We introduce the following quantities:

Definition 5 (Interaction functional and coefficients). The *interaction functional* is

$$Q_R(z) = -\sigma \left\langle \partial_{z/|z|} u_R(\cdot - z),\, u_R^{2p+1} \right\rangle_{L^2}.$$

The *minimum inter-peak distance* is $m_R = \min\{|z_j(R) - z_i(R)| : i \neq j\}$. The *interaction coefficients* are

$$\alpha_{ki} = \frac{Q_R(|z_k - z_i|)}{Q_R(m_R)}, \quad \text{satisfying } 0 \leq \alpha_{ki} \leq 1, \tag{2.5}$$

with $\alpha_{ki} = 1$ precisely when $|z_k - z_i| = m_R$ (nearest neighbors).

Definition 6 (Overall interaction strength). The *overall interaction strength* is

$$\chi = \frac{2\, Q_R(m_R)}{\|u_R\|_{L^2}^2\; R^4\, m_R} > 0. \tag{2.6}$$

This is a single positive scalar that captures the magnitude of the inter-peak interaction relative to the Hessian restoring force. It is *not* equal to any individual α_{ki}; rather, the force on peak i from peak k has magnitude $\alpha_{ki}\chi$.

The orthogonality condition from the Lyapunov–Schmidt projection yields, after the rescaling $s_i = Rz_i - x_0$ and passage to the limit $R \to 0$ with $y_i = \lim_{R\to 0} s_i/(Rm_R)$:

$$H_V(x_0)\, y_i = \chi \sum_{k \in N_i} \alpha_{ki}(y_k - y_i), \tag{2.7}$$

where $N_i = \{k : |y_k - y_i| = 1\}$ is the set of nearest neighbors and we have used $|y_k - y_i| = 1$ for nearest neighbors.

Corollary 7. *The peak positions satisfy* $\operatorname{span}\{y_1, \ldots, y_M\} \subseteq U$, *where* $U = \operatorname{span}\{\boldsymbol{r} \in \mathbb{R}^n : H_V(x_0)\boldsymbol{r} = \lambda \boldsymbol{r},\ \lambda < 0\}$ *is the unstable subspace of* $H_V(x_0)$.

Proof. Write $y_i = y_i^U + y_i^S$ where $y_i^U \in U$ and $y_i^S \in U^\perp$. Applying the projection P_S onto the stable subspace ($\lambda > 0$) to (2.7): $H_V P_S y_i = \chi \sum_{k \in N_i} \alpha_{ki} P_S(y_k - y_i)$. Since $H_V|_{U^\perp}$ has all positive eigenvalues and $\chi > 0$, $\alpha_{ki} \geq 0$, the right side is a convex combination of differences $P_S(y_k - y_i)$ scaled by positive factors. A discrete maximum principle argument shows $y_i^S = 0$ for all i. □

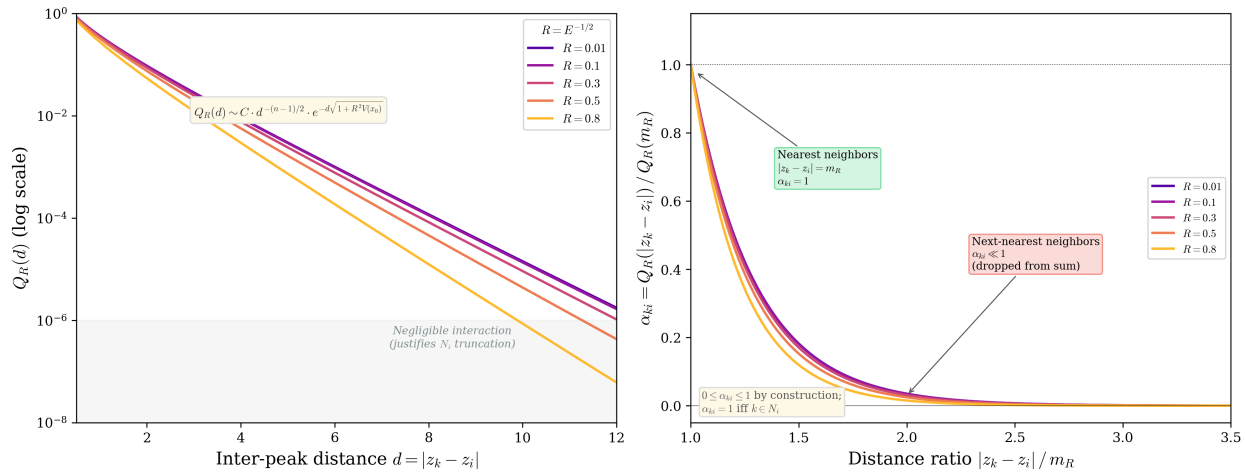

Figure 4: Left: Exponential decay of the interaction functional $Q_R(d) = -\sigma\langle \partial_{\hat{d}}\, u_R(\cdot - d),\, u_R^{2p+1}\rangle$ as a function of inter-peak distance d, illustrating that only nearest-neighbor interactions contribute at leading order. Right: The interaction coefficient $\alpha_{ki} = Q_R(|z_k - z_i|)/Q_R(m_R)$ as a function of the relative distance $|z_k - z_i|/m_R$, normalized to unity at the minimum distance.

3 Classification of Limiting Configurations

We now classify all admissible solutions of the algebraic system (2.7) for peaks lying in the plane spanned by two eigenvectors $\boldsymbol{r}_1, \boldsymbol{r}_2$ of $H_V(x_0)$, with eigenvalues $\lambda_1, \lambda_2 < 0$.

Theorem 8 (Classification of admissible multi-peak configurations)**.** *Under Hypotheses V1–V4, the admissible solutions of* (2.7) *with all* $\alpha_{ki} \ge 0$ *are:*

(a) ***Collinear*** *($M \ge 2$): All peaks aligned along a single eigenvector $\boldsymbol{r}_1$ of $H_V(x_0)$.*

(b) ***Isosceles triangle*** *($M = 3$): Peaks in the $(\boldsymbol{r}_1, \boldsymbol{r}_2)$-plane with $\alpha_{12} = \alpha_{13} = -\lambda_2/3$ and the eigenvalue constraint $\lambda_2 = 3\lambda_1$.*

(c) ***Equilateral triangle*** *($M = 3$): Peaks at unit mutual distance with $\alpha_{12} = \alpha_{13} = -\lambda_2/3$, $\alpha_{23} = \lambda_2/6 - \lambda_1/2$, requiring $|\lambda_2|/|\lambda_1| \ge 3$.*

(d) ***Rotational equilateral*** *($M = 3$, angle θ): The equilateral triangle rotated by angle θ relative to the eigenvector basis. The admissible (μ, θ) regions, where $\mu = |\lambda_2|/|\lambda_1|$, are given by Table 1.*

The proof of Theorem 8 occupies the remainder of this section.

3.1 Collinear configurations

Proof of Theorem 8(a). Let y_i be M connected collinear points along $\boldsymbol{r}_1$. For M even, set $y_{\pm i} = \pm(i - \frac{1}{2})\boldsymbol{r}_1$; for M odd, set $y_0 = 0$, $y_{\pm i} = \pm i\, \boldsymbol{r}_1$. The nearest-neighbor constraint gives

$$H_V\, y_i = (\alpha_{i,i+1} + \alpha_{i,i-1})\boldsymbol{r}_1$$

with $\alpha_{i,i\pm1} \ge 0$ determined recursively. Since $\boldsymbol{r}_1$ is an eigenvector with $\lambda_1 < 0$, each equation reduces to a scalar relation $\lambda_1 c_i = \chi(\alpha_{i,i+1} + \alpha_{i,i-1})$ with c_i the coefficient of y_i along $\boldsymbol{r}_1$. The non-negativity $\alpha \ge 0$ is guaranteed by $\lambda_1 < 0$ and $\chi > 0$. □

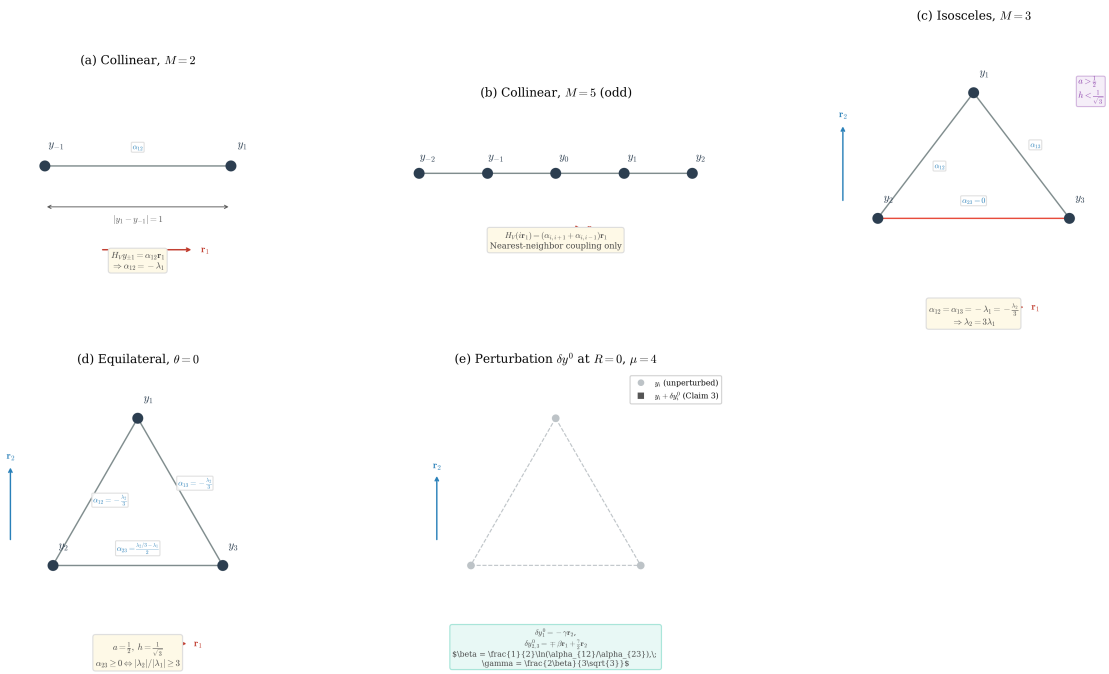

Figure 5: Representative peak configurations classified in Theorem 8. From left to right: collinear arrangements with $M = 2$ and $M = 5$ peaks along a single eigenvector; isosceles triangle ($M = 3$) requiring $\lambda_2 = 3\lambda_1$; equilateral triangle ($M = 3$) requiring $|\lambda_2|/|\lambda_1| \geq 3$; and equilateral triangle with rotational degree of freedom θ, with a schematic perturbation δy.

3.2 Isosceles triangle

Proof of Theorem 8(b). Let $y_1 = (0, h)^\intercal$, $y_2 = (-a, -h/2)^\intercal$, $y_3 = (a, -h/2)^\intercal$ with $2a > 1$ (so $|y_2 - y_3| = 2a > 1$ and the nearest-neighbor pairs are $(1, 2)$ and $(1, 3)$ only) and the geometric constraint $a^2 + (3h/2)^2 = 1$.

From the equation for y_1:

$$\begin{aligned} H_V y_1 &= \alpha_{12}(y_2 - y_1) + \alpha_{13}(y_3 - y_1) \\ (0, \lambda_2 h)^\intercal &= \big(a(\alpha_{13} - \alpha_{12}),\ -\tfrac{3h}{2}(\alpha_{13} + \alpha_{12})\big)^\intercal, \end{aligned}$$

which gives $\alpha_{12} = \alpha_{13} = -\lambda_2/3$.

From the equation for y_2 (with only y_1 as nearest neighbor): $H_V y_2 = \alpha_{12}(y_1 - y_2)$, yielding component-wise $-a\lambda_1 = a\alpha_{12}$ and $-(h/2)\lambda_2 = (3h/2)\alpha_{12}$. The first gives $\alpha_{12} = -\lambda_1$ and the second $\alpha_{12} = -\lambda_2/3$, hence $\lambda_2 = 3\lambda_1$. □

3.3 Equilateral triangle

Proof of Theorem 8(c). Set $2a = 1$, giving $a = 1/2$, $h = 1/\sqrt{3}$, and $|y_k - y_i| = 1$ for all pairs. All three pairs are nearest neighbors. From y_1: $\alpha_{12} = \alpha_{13} = -\lambda_2/3$ as before. From y_2 (with both y_1 and y_3 as neighbors):

$$\begin{aligned} H_V y_2 &= \alpha_{12}(y_1 - y_2) + \alpha_{23}(y_3 - y_2), \\ (-a\lambda_1,\ -\tfrac{h}{2}\lambda_2)^\intercal &= (a\alpha_{12} + 2a\alpha_{23},\ \tfrac{3h}{2}\alpha_{12})^\intercal, \end{aligned}$$

giving $\alpha_{23} = \frac{\lambda_2}{6} - \frac{\lambda_1}{2}$. The constraint $\alpha_{23} \geq 0$ requires $\lambda_2 \geq 3\lambda_1$, i.e., $|\lambda_2|/|\lambda_1| \geq 3$. □

3.4 Rotational equilateral triangle

Proof of Theorem 8(d). Let $R(\theta)$ be the 2×2 rotation matrix by angle θ and set $y_i = R(\theta) y_i'$ where y_i' are the canonical equilateral positions. The conjugated Hessian $\tilde{H} = R(-\theta) H_V R(\theta)$ has entries

$$\tilde{H} = \begin{pmatrix} \lambda_1 \cos^2\theta + \lambda_2 \sin^2\theta & (\lambda_2 - \lambda_1)\sin\theta\cos\theta \\ (\lambda_2 - \lambda_1)\sin\theta\cos\theta & \lambda_2\cos^2\theta + \lambda_1\sin^2\theta \end{pmatrix}.$$

Solving $\tilde{H}\, y_i' = \sum_{k \in N_i} \alpha_{ki}(y_k' - y_i')$ for each vertex yields the θ-dependent coefficients:

$$\alpha_{12}(\theta) = \tfrac{1}{3}\big[\lambda_2(-\sqrt{3}\sin\theta - \cos\theta)\cos\theta + \lambda_1(-\sin\theta + \sqrt{3}\cos\theta)\sin\theta\big], \tag{3.1}$$

$$\alpha_{13}(\theta) = \tfrac{1}{3}\big[\lambda_2(\sqrt{3}\sin\theta - \cos\theta)\cos\theta - \lambda_1(\sin\theta + \sqrt{3}\cos\theta)\sin\theta\big], \tag{3.2}$$

$$\alpha_{23}(\theta) = \tfrac{1}{6}\big[\lambda_2(2\cos 2\theta - 1) - \lambda_1(2\cos 2\theta + 1)\big]. \tag{3.3}$$

□

3.5 Admissibility analysis

The constraints $\alpha_{12}, \alpha_{13}, \alpha_{23} \geq 0$ with $\mu = |\lambda_2|/|\lambda_1|$ reduce via double-angle identities to

$$(\mu + 1) + 2(\mu - 1)\cos\phi \geq 0, \quad \phi \in \big\{2\theta - \tfrac{\pi}{3},\ 2\theta,\ 2\theta + \tfrac{\pi}{3}\big\}. \tag{3.4}$$

Definition 9. Define $\nu(\mu) = \frac{\mu+1}{2|\mu-1|}$ and $\varphi(\mu) = \arccos(\nu(\mu))$.

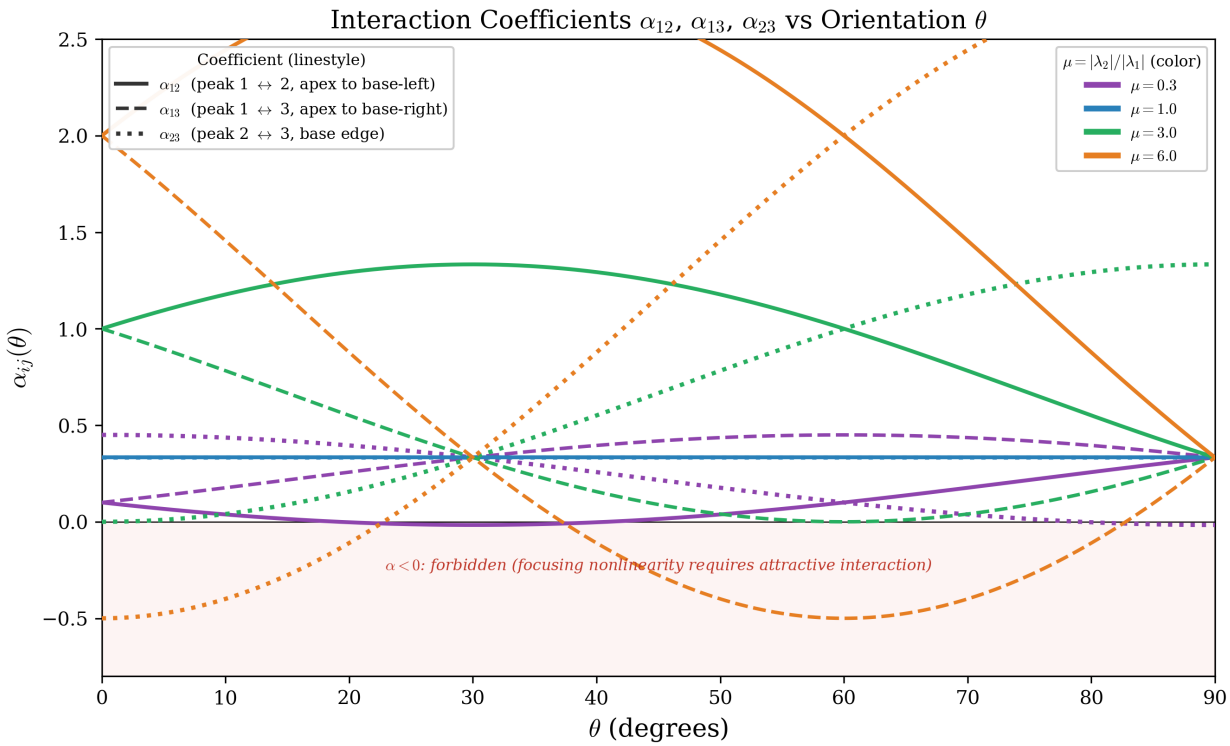

Figure 6: Interaction coefficients $\alpha_{12}(\theta)$, $\alpha_{13}(\theta)$, $\alpha_{23}(\theta)$ as functions of the rotation angle $\theta \in [0, \pi/2]$ for four representative eigenvalue ratios $\mu \in \{0.3, 1.0, 3.0, 6.0\}$. A configuration is admissible at angle θ if and only if all three coefficients are non-negative. For $\mu = 1$ (isotropic potential) all orientations are admissible. For $\mu = 6 > 3$, the admissible θ-range breaks into two disjoint intervals.

Proposition 10 (Admissibility regions)**.** *The admissible* (μ, θ) *regions are:*

(i) If $\frac{1}{3} \le \mu \le 3$*, then* $\nu(\mu) \ge 1$ *and* $\varphi = 0$*: all* $\theta \in [0, \pi/2]$ *are admissible.*

(ii) If $0 < \mu < \frac{1}{3}$*, then* $\nu(\mu) \in (\frac{1}{2}, 1)$ *and the admissible angles are* $\theta \in [0, \frac{\pi}{6} - \frac{\varphi}{2}] \cup [\frac{\pi}{6} + \frac{\varphi}{2}, \frac{\pi}{2} - \frac{\varphi}{2}]$.

(iii) If $\mu > 3$*, then* $\nu(\mu) \in (\frac{1}{2}, 1)$ *and the admissible angles are* $\theta \in [\frac{\varphi}{2}, \frac{\pi}{3} - \frac{\varphi}{2}] \cup [\frac{\pi}{3} + \frac{\varphi}{2}, \frac{\pi}{2}]$.

Table 1: Summary of admissible θ-ranges as a function of $\mu = |\lambda_2|/|\lambda_1|$.

μ-Range	θ-Range
$0 < \mu < \frac{1}{3}$	$\left[0,\ \frac{\pi}{6} - \frac{\varphi(\mu)}{2}\right] \cup \left[\frac{\pi}{6} + \frac{\varphi(\mu)}{2},\ \frac{\pi}{2} - \frac{\varphi(\mu)}{2}\right]$
[8pt] $\frac{1}{3} \le \mu \le 3$	$\left[0,\ \frac{\pi}{2}\right]$
[8pt] $\mu > 3$	$\left[\frac{\varphi(\mu)}{2},\ \frac{\pi}{3} - \frac{\varphi(\mu)}{2}\right] \cup \left[\frac{\pi}{3} + \frac{\varphi(\mu)}{2},\ \frac{\pi}{2}\right]$

4 The Perturbation System

We now extend the limiting configurations ($R = 0$) to exact solutions at finite energy ($R > 0$). We focus on the equilateral triangle at $\theta = 0$.

4.1 Formulation

Write $\tilde{y}_i = \tilde{m}_R\, y_i + \delta y_i$ with $\delta y_i / \tilde{m}_R \to 0$ as $R \to 0$, where $\delta y_i \in \operatorname{span}\{\boldsymbol{r}_1, \boldsymbol{r}_2\}$.

Definition 11 (Perturbation system)**.** Define

$$F_i(\delta y, R) = H_V y_i + H_V \frac{\delta y_i}{\tilde{m}_R} - \sum_{k \in N_i} \frac{2Q_R\big(|\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|\big)}{\|u_R\|_{L^2}^2\ R^4\, |\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|} \Big(y_k - y_i + \frac{\delta y_k - \delta y_i}{\tilde{m}_R}\Big). \tag{4.1}$$

Extending by continuity to $R = 0$:

$$F_i(\delta y, 0) = H_V y_i - \chi \sum_{k \in N_i} e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle} (y_k - y_i), \tag{4.2}$$

where χ is the overall interaction strength (Definition 6), and the perturbed interaction coefficients are $\tilde{\alpha}_{ki} = e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle}$, satisfying $\tilde{\alpha}_{ki}\chi \to \alpha_{ki}\chi$ as $\delta y \to 0$. Note that the constraint $\max_k \alpha_{ki} = 1$ from (2.5) normalizes χ to equal the total force from a nearest-neighbor pair at minimum distance.

The Fréchet derivative is

$$D_{\delta y} F_i(\delta y, 0)[w] = \chi \sum_{k \in N_i} \langle w_k - w_i, y_k - y_i \rangle\, e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle} (y_k - y_i). \tag{4.3}$$

4.2 Explicit solution at $R = 0$

Proposition 12 (Explicit solution)**.** *Define* $\beta = \frac{1}{2} \ln(\alpha_{12}/\alpha_{23})$ *and* $\gamma = \frac{2\beta}{3\sqrt{3}}$*, where* $\alpha_{12} = -\lambda_2/3$ *and* $\alpha_{23} = \lambda_2/6 - \lambda_1/2$ *are the unperturbed coefficients. Then*

$$\delta y_1^0 = -\gamma\, \boldsymbol{r}_2, \qquad \delta y_{2,3}^0 = \mp\beta\, \boldsymbol{r}_1 + \tfrac{\gamma}{2}\, \boldsymbol{r}_2 \tag{4.4}$$

solves $F_i(\delta y, 0) = 0$ *for* $i = 1, 2, 3$.

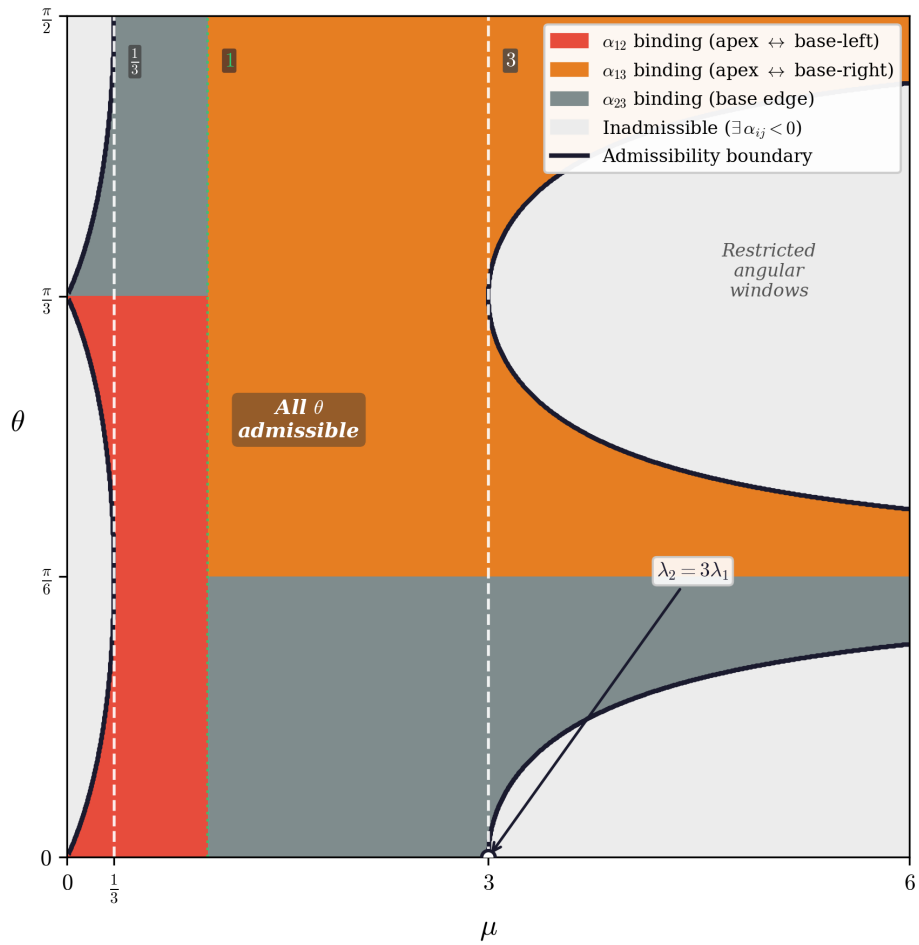

Figure 7: Admissible region (shaded) in the (μ, θ) parameter space for equilateral triangle configurations with rotational degree of freedom. The dashed vertical lines mark $\mu = 1/3$ and $\mu = 3$, the boundaries of the fully admissible strip. For μ outside this strip, the constraint $\alpha_{ki} \geq 0$ excludes angular windows whose width is controlled by $\varphi(\mu) = \arccos(\nu(\mu))$. The boundary curves correspond to the binding constraints where exactly one $\alpha_{ki} = 0$.

Proof. The perturbed interaction coefficients at δy^0 are (computed by direct substitution):

$$\tilde{\alpha}_{12} = \tilde{\alpha}_{13} = e^{-\langle y_2 - y_1, \delta y_2^0 - \delta y_1^0 \rangle} = 1,$$
$$\tilde{\alpha}_{23} = \tilde{\alpha}_{32} = e^{-\langle y_3 - y_2, \delta y_3^0 - \delta y_2^0 \rangle} = \frac{\alpha_{23}}{\alpha_{12}}.$$

The first follows from the fact that $\langle y_2 - y_1, \delta y_2^0 - \delta y_1^0 \rangle = 0$ (which one verifies by substituting the explicit coordinates). The second uses $\langle y_3 - y_2, \delta y_3^0 - \delta y_2^0 \rangle = 2\beta = \ln(\alpha_{12}/\alpha_{23})$.

Setting $\chi = \alpha_{12}$ (the nearest-neighbor coefficient), the system $F_i(\delta y^0, 0) = 0$ becomes

$$H_V y_i = \alpha_{12}\big[\tilde{\alpha}_{i1}(y_1 - y_i) + \tilde{\alpha}_{i,\text{other}}(y_{\text{other}} - y_i)\big]$$

for each i, which reduces to the original algebraic system (2.7). Substituting the known values and simplifying component-wise confirms $F_i = 0$ for each i.

We also verify the center-of-mass condition: $\delta y_1^0 + \delta y_2^0 + \delta y_3^0 = (0, 0)$, which is immediate from the definitions. □

4.3 Kernel of the linearization

Proposition 13 (Nontrivial kernel)**.** *The linearization $D_{\delta y} F_i(\delta y^0, 0)$ has a four-dimensional kernel*

$$\Omega = \{(w_1, w_2, w_3) \in (\mathbb{R}^2)^3 : \langle w_k - w_i, y_k - y_i \rangle = 0 \ \ \forall\, i \neq k\}.$$

This kernel contains the two-dimensional translational subspace $\{(v, v, v) : v \in \mathbb{R}^2\}$ and two additional directions, including the rotational mode $w_{\text{rot}} = \frac{d}{d\theta} y_i(\theta)\big|_{\theta=0}$.

Proof. The kernel condition $\langle w_k - w_i, y_k - y_i \rangle = 0$ for all neighbor pairs gives three scalar constraints on $w \in \mathbb{R}^6$. Since $y_3 - y_2 = (y_3 - y_1) - (y_2 - y_1)$, only two constraints are independent, so $\dim \Omega \geq 6 - 2 = 4$.

To see that $\Omega \supseteq \{(v, v, v) : v \in \mathbb{R}^2\}$: if $w_1 = w_2 = w_3 = v$, then $w_k - w_i = 0$, so all inner products vanish. This is a 2D subspace.

The quadratic form argument shows the inclusion is an equality: define

$$\Phi = \sum_i w_i \cdot D_{\delta y} F_i(\delta y^0, 0)[w] = -\chi \sum_{\{i,k\} \in \text{edges}} \alpha_{ik} \big(\langle w_k - w_i, y_k - y_i \rangle\big)^2.$$

Since $\Phi \leq 0$ and $\Phi = 0$ implies each summand vanishes (as $\alpha_{ik} > 0$ for all edges of the equilateral triangle), the kernel is precisely Ω. Computing explicitly, $\dim \Omega = 4$.

The rotational mode $w_{\text{rot}} = (\frac{d}{d\theta} y_i(\theta)|_{\theta=0})_{i=1,2,3} = (-h, 0;\ \frac{h}{2}, -a;\ \frac{h}{2}, a)$ lies in Ω since $\langle w_{\text{rot},k} - w_{\text{rot},i}, y_k - y_i \rangle = 0$ for all pairs (verified by direct computation). □

Remark 14. The four kernel dimensions decompose as: 2 from translations, 1 from rotation, and 1 additional "breathing" mode. The gauge-fixing must remove at least 2 directions to allow the implicit function theorem to apply on the complement.

5 Kernel Resolution and Gauge-Fixing

The four-dimensional kernel Ω prevents a direct application of the implicit function theorem (IFT) to continue solutions from $R = 0$ to $R > 0$. Following the iterative kernel decomposition methodology of [7], we decompose the reduced system into projections perpendicular and parallel to Ω.

5.1 Projection decomposition

For $w \in \Omega$, define

$$P_w F(\delta y, R) = \sum_i \langle w_i, F_i(\delta y, R)\rangle.$$

We decompose F into its perpendicular projection $P_\perp F$ (a vector-valued system on $\Omega^\perp$) and parallel projections $P_w F$ for w ranging over a basis of Ω.

At $(\delta y^0, 0)$, the linearizations are:

$$D_{\delta y} P_\perp F(\delta y^0, 0)[\delta w] = P_\perp \sum_{k \in N_i} \langle \delta w_k - \delta w_i, y_k - y_i\rangle\, \tilde{\alpha}_{ik}(0)\, \chi\,(y_k - y_i), \tag{5.1}$$

$$D_{\delta y} P_w F(\delta y^0, 0)[\delta w] = \sum_i \Big[\langle w_i, H_V \delta w_i\rangle + \sum_{k \in N_i} \tilde{\alpha}_{ki}(0)\big(\langle w_i, \delta y_k^0 - \delta y_i^0\rangle \langle y_k - y_i, \delta w_k - \delta w_i\rangle - \langle w_i, \delta w_k - \delta w_i\rangle\big)\Big]. \tag{5.2}$$

5.2 Gauge-fixing constraints

To break the translational and rotational symmetries, we impose:

(G1) $\delta y_1 \perp \boldsymbol{r}_1$ (the first peak has no displacement along $\boldsymbol{r}_1$),

(G2) $(\delta y_3 - \delta y_2) \parallel \boldsymbol{r}_1$ (the base of the triangle displaces only along $\boldsymbol{r}_1$).

These two scalar constraints remove 2 of the 4 kernel directions (one translation component and the rotation mode), leaving a 2-dimensional residual kernel corresponding to the remaining translation and the breathing mode.

5.3 Rotational equivariance at $R = 0$

Proposition 15 (Family structure at $R = 0$)**.** *The one-parameter family of equilateral triangle configurations $y_i(\theta) = R(\theta) y_i'$, $\theta \in [\theta_{\min}, \theta_{\max}]$ (the admissible θ-range from Proposition 10), generates a corresponding family of solutions $\delta y^0(\theta)$ of the system $F_i(\delta y, 0) = 0$.*

Proof. For each admissible θ, the base configuration $y_i(\theta)$ satisfies the algebraic system (2.7) with θ-dependent coefficients $\alpha_{12}(\theta), \alpha_{13}(\theta), \alpha_{23}(\theta)$ from (3.1)–(3.3). The explicit solution construction of Proposition 12 applies at each θ, with $\beta(\theta) = \frac{1}{2}\ln(\alpha_{12}(\theta)/\alpha_{23}(\theta))$ and $\gamma(\theta) = 2\beta(\theta)/(3\sqrt{3})$. The resulting $\delta y^0(\theta)$ is smooth in θ since the $\alpha_{ki}(\theta)$ are smooth and positive in the admissible range. □

Corollary 16. *The tangent vector $\frac{d}{d\theta}\delta y^0(\theta)$ at each θ lies in the kernel $\Omega(\theta)$ of $D_{\delta y} F_i(\delta y^0(\theta), 0)$. This is the rotational mode $w_{\mathrm{rot}}(\theta)$.*

Proof. Differentiating $F_i(\delta y^0(\theta), 0) = 0$ with respect to θ gives $D_{\delta y} F_i\big[\frac{d}{d\theta}\delta y^0\big] + \partial_\theta F_i = 0$. Since $\partial_\theta F_i$ encodes the change in the base configuration (which is already accounted for), the tangent direction lies in the kernel. □

5.4 Extension to $R > 0$: invertibility of the gauge-fixed system

Conjecture 17 (Gauge-fixed invertibility)**.** *Under gauge-fixing conditions (G1)–(G2), the projected system*

$$P_\perp F(\delta y, R) = 0, \qquad P_w F(\delta y, R) = 0 \quad (w \in \Omega \setminus \mathrm{span}\{w_{\mathrm{trans}}, w_{\mathrm{rot}}\})$$

restricted to the gauge-fixed subspace $\{\delta y : \delta y_1 \perp \boldsymbol{r}_1,\ (\delta y_3 - \delta y_2) \parallel \boldsymbol{r}_1\}$ has an invertible linearization at $(\delta y^0, 0)$, and hence the IFT yields a unique smooth branch $\delta y(R)$ for $R \in [0, \varepsilon)$.

The conjecture is supported by the following partial result:

Proposition 18. *The constraint $\sum_i F_i(\delta y, R) = 0$ is automatically satisfied and recovers the reduced system (2.7) in the limit $R \to 0$. The gauge-fixing conditions (G1)–(G2) are compatible with the solution δy^0 (which satisfies them by construction).*

Proof. Summing F_i over i: each interaction term $\alpha_{ik}(y_k - y_i)$ appears once with sign $+$ (from peak i) and once with sign $-$ (from peak k), so the interaction terms cancel in pairs. What remains is

$$\sum_i F_i = \sum_i H_V \frac{\tilde{y}_i}{\tilde{m}_R} - \sum_i \sum_{k \in N_i} \frac{2Q_R(\cdots)}{\cdots} \cdot \frac{\tilde{y}_k - \tilde{y}_i}{\tilde{m}_R} = \sum_i \Big[H_V \frac{\tilde{y}_i}{\tilde{m}_R} - \sum_{k \in N_i} \cdots \Big] = 0,$$

where the last equality follows from the fact that each term in the sum is precisely the reduced system (2.7) applied to peak i.

For compatibility: $\delta y_1^0 = -\gamma \boldsymbol{r}_2$ satisfies $\delta y_1^0 \perp \boldsymbol{r}_1$ since it has no $\boldsymbol{r}_1$-component. And $\delta y_3^0 - \delta y_2^0 = 2\beta \boldsymbol{r}_1 \parallel \boldsymbol{r}_1$. □

Remark 19. The full resolution of Conjecture 17 requires an explicit computation of the 4×4 matrix $D_{\delta y}(P_\perp F \oplus P_\parallel F)$ restricted to the gauge-fixed 4-dimensional subspace, and showing its 2×2 restriction to $\Omega^\perp$ within the gauge-fixed space is nonsingular. For the isotropic case $\mu = 1$ ($\lambda_1 = \lambda_2$), the full continuous rotational symmetry $H_V = \lambda I$ implies equivariance $F_i(R_\omega \delta y, R) = R_\omega F_i(\delta y, R)$ for all ω, and the IFT applies on the quotient by the rotation group. For anisotropic $\mu \neq 1$, the equivariance is broken but the family structure (Proposition 15) provides the necessary tangent-space information.

6 Conclusion and Open Problems

We have carried out a complete finite-dimensional study of multi-peak ground states for the nonlinear Schrödinger equation

$$(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0, \qquad E \to \infty,$$

under Hypotheses V1–V4. Our main results are:

1. **Classification** (Theorem 8). All admissible multi-peak configurations near a non-degenerate local maximum of V are classified: collinear, isosceles triangle ($\lambda_2 = 3\lambda_1$), equilateral triangle ($|\lambda_2/\lambda_1| \geq 3$), and equilateral with rotational freedom (Table 1).

2. **Perturbation system** (Propositions 12–13). The system $F_i(\delta y, R) = 0$ admits an explicit solution δy^0 at $R = 0$, and the linearization has a four-dimensional kernel Ω (translations + rotation + breathing mode).

3. **Kernel resolution strategy** (Section 5). Gauge-fixing conditions remove the translational and rotational freedom. The family structure (Proposition 15) provides the tangent information, and Conjecture 17 identifies the remaining technical step.

Open problems. The following concrete, finite-dimensional problems remain:

1. **Prove Conjecture 17** by computing the linearized operator on the gauge-fixed subspace explicitly (a 4×4 matrix computation) and verifying nonsingularity for $\mu > 3$.

2. **Extend to all admissible θ.** The current perturbation analysis is carried out at $\theta = 0$. Proposition 15 shows the family exists at $R = 0$; extending to $R > 0$ for all admissible θ simultaneously requires uniform estimates.

3. **Stability.** Multi-peak branches with $M \geq 2$ have at least two negative directions in the linearized PDE operator [2], implying orbital instability. Adapting the asymptotic stability techniques of [8, 6, 4] to multi-peak states remains open.

Once Conjecture 17 is resolved, the Lyapunov–Schmidt machinery of [7] lifts the finite-dimensional solutions to exact multi-peak ground states of (1.1) for all sufficiently large energies E.

References

[1] Andreas Floer and Alan Weinstein. Nonspreading wave packets for the cubic schrödinger equation with a bounded potential. *Journal of functional analysis*, 69(3):397–408, 1986. 1, 2.1, 1

[2] Manoussos Grillakis, Jalal Shatah, and Walter Strauss. Stability theory of solitary waves in the presence of symmetry, i. *Journal of functional analysis*, 74(1):160–197, 1987. 1, 3

[3] Manoussos Grillakis, Jalal Shatah, and Walter Strauss. Stability theory of solitary waves in the presence of symmetry, ii. *Journal of functional analysis*, 94(2):308–348, 1990. 1

[4] E. Kirr. Long time dynamics and coherent states in nonlinear wave equations, 2016. 3

[5] E. Kirr, P. G. Kevrekidis, and D. E. Pelinovsky. Symmetry-breaking bifurcation in the nonlinear schrödinger equation with symmetric potentials, October 2011. 1, 2.1

[6] E. Kirr and O. Mizrak. Asymptotic stability of ground states in 3d nonlinear schrödinger equation including subcritical cases, 2008. 1, 3

[7] E. Kirr and V. Natarajan. The global bifurcation picture for ground states in nonlinear schrödinger equations, 2018. (document), 1, 1, 2.1, 2.2, 2, 3, 4, 5, 6

[8] E. Kirr and A. Zarnescu. On the asymptotic stability of bound states in 2d cubic schrödinger equation, March 2007. 1, 3

[9] P.L. Lions. The concentration-compactness principle in the calculus of variations. the locally compact case, part 1. *Annales de l'Institut Henri Poincaré. Analyse non linéaire*, 1(2):109–145, 1984. 1, 1, 2.2

[10] P.L. Lions. The concentration-compactness principle in the calculus of variations. the locally compact case, part 2. *Annales de l'Institut Henri Poincaré. Analyse non linéaire*, 1(4):223–283, 1984. 1

A Notation Reference

For the reader's convenience, we collect the principal notation.

Symbol	Description
ψ_E	Solution of (1.1) at energy E
V, $H_V(x_0)$	External potential and its Hessian at x_0
λ_i, $\boldsymbol{r}_i$	Eigenvalues/eigenvectors of $H_V(x_0)$; $\lambda_i < 0$
$\mu = \lvert\lambda_2\rvert/\lvert\lambda_1\rvert$	Eigenvalue ratio
$R = E^{-1/2}$	Rescaling parameter; $R \to 0$ as $E \to \infty$
$u_E(x) = E^{-1/(2p)}\psi_E(E^{-1/2}x)$	Rescaled solution
$Q_R(z)$	Interaction functional (Definition 5)
α_{ki}	Interaction coefficient (Definition 5); $0 \leq \alpha_{ki} \leq 1$
χ	Overall interaction strength (Definition 6)
m_R	Minimum inter-peak distance
$y_i = \lim s_i/(Rm_R)$	Limiting peak positions
δy_i	Perturbation of peak i; $\delta y_i \in \operatorname{span}\{\boldsymbol{r}_1, \boldsymbol{r}_2\}$
$F_i(\delta y, R)$	Perturbation system (Definition 11)
$\tilde{\alpha}_{ki} = e^{-\langle y_k - y_i, \delta y_k - \delta y_i\rangle}$	Perturbed interaction coefficient
Ω	Kernel of $D_{\delta y}F_i(\delta y^0, 0)$ (Proposition 13)
θ	Rotation angle of equilateral triangle
$\nu(\mu), \varphi(\mu)$	Admissibility functions (Definition 9)